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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 29th June Evening Shift

JEE Main Question Paper with Answer Key

29 Jun 2022	90	20	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	23 questions
Mathematics	20 questions
Physics	27 questions
Out of syllabus	20 questions
Answer key	at the end

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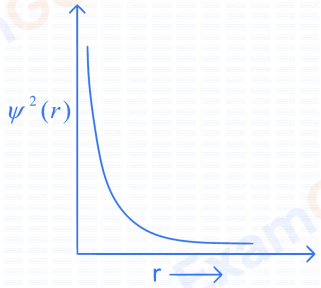
Chemistry 23 questions

Q1. MCQ (Single Correct Answer)

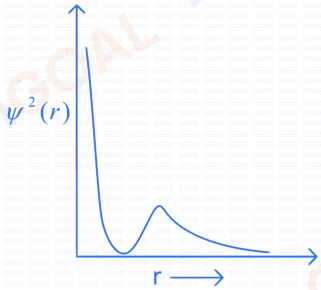
Open solution

Which of the following is the correct plot for the probability density ψ^2 (r) as a function of distance 'r' of the electron from the nucleus for 2s orbital?

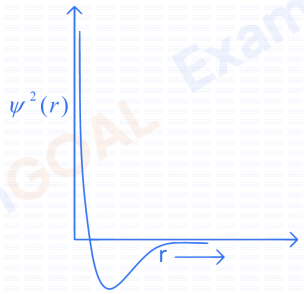
(A)



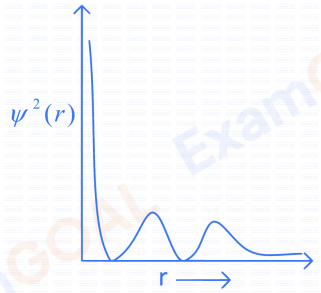
(B)



(C)



(D)



Q2. MCQ (Single Correct Answer)

Open solution

Consider the species CH_4, NH_4^+ and BH_4^-. Choose the correct option with respect to these species.

(A) They are isoelectronic and only two have tetrahedral structures.

(B) They are isoelectronic and all have tetrahedral structures.

(C) Only two are isoelectronic and all have tetrahedral structures.

(D) Only two are isoelectronic and only two have tetrahedral structures.

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Q3. MCQ (Single Correct Answer)

4.0 moles of argon and 5.0 moles of PCl_5 are introduced into an evacuated flask of 100 litre capacity at 610 K. The system is allowed to equilibrate. At equilibrium, the total pressure of mixture was found to be 6.0 atm. The K_p for the reaction is :

[Given : R = 0.082 L atm K^{-1} mol^{-1}]

(A) 2.25 (B) 6.24 (C) 12.13 (D) 15.24

Open solution

Q4. MCQ (Single Correct Answer)

Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : The first ionization enthalpy for oxygen is lower than that of nitrogen.

Reason R : The four electrons in 2p orbitals of oxygen experience more electron-electron repulsion.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both A and R are correct and R is the correct explanation of A.

(B) Both A and R are correct but R is NOT the correct explanation of A.

(C) A is correct but R is not correct.

(D) A is not correct but R is correct.

Open solution

Q5. MCQ (Single Correct Answer)

Given below are two statements.

- Statement I : In $CuSO_4 . 5H_2O$, Cu-O bonds are present.
- Statement II : In $CuSO_4 . 5H_2O$, ligands coordinating with Cu(II) ion are O-and S-based ligands.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Open solution

Q6. MCQ (Single Correct Answer)

Number of lone pair(s) of electrons on central atom and the shape BrF_3 molecule respectively, are

(A) 0, triangular planar. (B) 1, pyramidal.

(C) 2, bent T-shape. (D) 1, bent T-shape.

Open solution

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Q7. MCQ (Single Correct Answer)

Open solution

Sulphur dioxide is one of the components of polluted air. SO_2 is also a major contributor to acid rain. The correct and complete reaction to represent acid rain caused by SO_2 is :

(A) $2SO_2 + O_2 \rightarrow 2SO_3$

(B) $SO_2 + O_3 \rightarrow SO_3 + O_2$

(C) $SO_2 + H_2O_2 \rightarrow H_2SO_4$

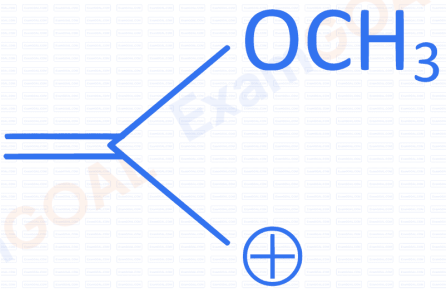
(D) $2SO_2 + O_2 + 2H_2O \rightarrow 2H_2SO_4$

Q8. MCQ (Single Correct Answer)

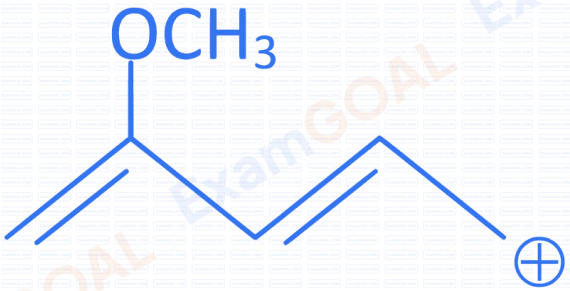
Open solution

Which of the following carbocations is most stable?

(A)



(B)



(C)



(D)



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Q9. MCQ (Single Correct Answer)



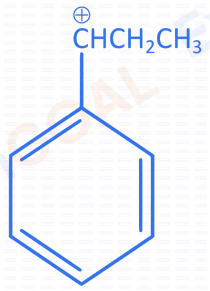
Open solution

The stable carbocation formed in the above reaction is

(A) $CH_3CH_2\overset{\oplus}{C}H_2$

(B) $CH_3\overset{\oplus}{C}H_2$

(C) $CH_3-\overset{\oplus}{C}H-CH_3$

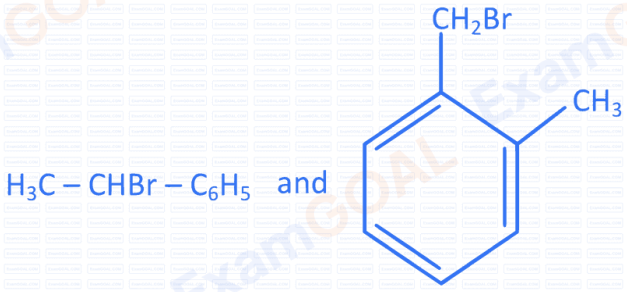
(D) 

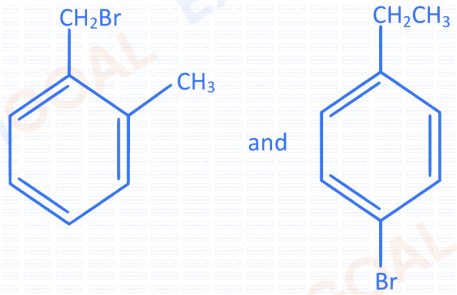
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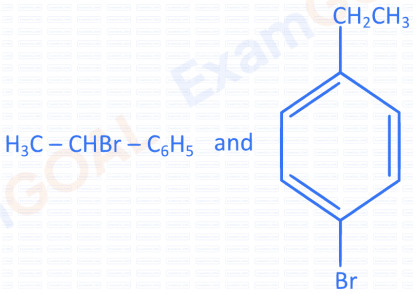
Q10. MCQ (Single Correct Answer)

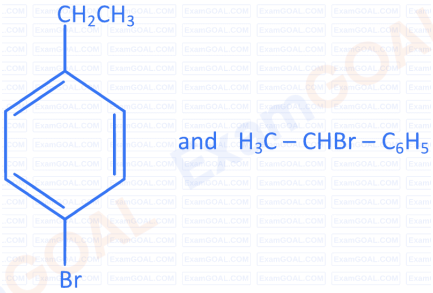
Open solution

Two isomers (A) and (B) with Molar mass 184 g/mol and elemental composition C, 52.2%; H, 4.9% and Br 42.9% gave benzoic acid and p-bromobenzoic acid, respectively on oxidation with $KMnO_4$. Isomer 'A' is optically active and gives a pale yellow precipitate when warmed with alcoholic $AgNO_3$. Isomer 'A' and 'B' are, respectively

(A) 

(B) 

(C) 

(D) 

Q11. MCQ (Single Correct Answer)

Open solution

In Friedel-Crafts alkylation of aniline, one gets

(A) alkylated product with ortho and para substitution.

(B) secondary amine after acidic treatment.

(C) an amide product.

(D) positively charged nitrogen at benzene ring.

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Q12. MCQ (Single Correct Answer)

The structure of protein that is unaffected by heating is

(A) secondary structure
(B) tertiary structure
(C) primary structure
(D) quaternary structure

Open solution

Q13. MCQ (Single Correct Answer)

A white precipitate was formed when $BaCl_2$ was added to water extract of an inorganic salt. Further, a gas 'X' with characteristic odour was released when the formed white precipitate was dissolved in dilute HCl. The anion present in the inorganic salt is

(A) I^-
(B) SO_3^{2-}
(C) S^{2-}
(D) NO_2^-

Open solution

Q14. Numerical

A box contains 0.90 g of liquid water in equilibrium with water vapour at 27°C. The equilibrium vapour pressure of water at 27°C is 32.0 Torr. When the volume of the box is increased, some of the liquid water evaporates to maintain the equilibrium pressure. If all the liquid water evaporates, then the volume of the box must be _____ litre. [nearest integer]

(Given : R = 0.082 L atm K^{-1} mol^{-1})

(Ignore the volume of the liquid water and assume water vapours behave as an ideal gas.)

Open solution

Q15. Numerical

2.2 g of nitrous oxide (N_2O) gas is cooled at a constant pressure of 1 atm from 310 K to 270 K causing the compression of the gas from 217.1 mL to 167.75 mL. The change in internal energy of the process, ΔU is '$-x$' J. The value of 'x' is _____. [nearest integer]

(Given : atomic mass of N = 14 g mol^{-1} and of O = 16 g mol^{-1}. Molar heat capacity of N_2O is 100 J K^{-1} mol^{-1})

Open solution

Q16. Numerical

Elevation in boiling point for 1.5 molal solution of glucose in water is 4 K. The depression in freezing point for 4.5 molal solution of glucose in water is 4 K. The ratio of molal elevation constant to molal depression constant (K_b/K_f) is _____.

Open solution

Q17. Numerical

The cell potential for the given cell at 298 K

Pt| H_2 (g, 1 bar) | H^+ (aq) || Cu^{2+} (aq) | Cu(s)

is 0.31 V. The pH of the acidic solution is found to be 3, whereas the concentration of Cu^{2+} is 10^{-x} M. The value of x is _____.

(Given : $E^{\Theta}_{Cu^{2+}/Cu}$ = 0.34 V and $\frac{2.303\,RT}{F}$ = 0.06 V)

Open solution

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Q18. Numerical

The equation

$k = (6.5 \times 10^{12} s^{-1}) e^{-26000K/T}$

is followed for the decomposition of compound A. The activation energy for the reaction is _____ kJ mol^{-1}. [nearest integer]

(Given : R = 8.314 J K^{-1} mol^{-1})

Open solution

Q19. Numerical

Spin only magnetic moment of $[MnBr_6]^{4-}$ is _____ B.M. (round off to the closest integer)

Open solution

Q20. Numerical

For the reaction given below :

$CoCl_3 . xNH_3 + AgNO_3 (aq) \rightarrow$

If two equivalents of AgCl precipitate out, then the value of x will be _____.

Open solution

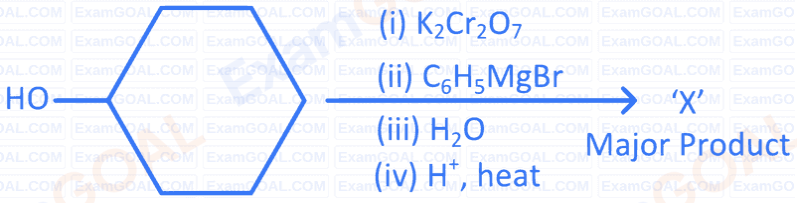
Q21. Numerical

The number of chiral alcohol(s) with molecular formula $C_4H_{10}O$ is _____.

Open solution

Q22. Numerical

In the given reaction,



the number of sp^2 hybridised carbon(s) in compound 'X' is _____.

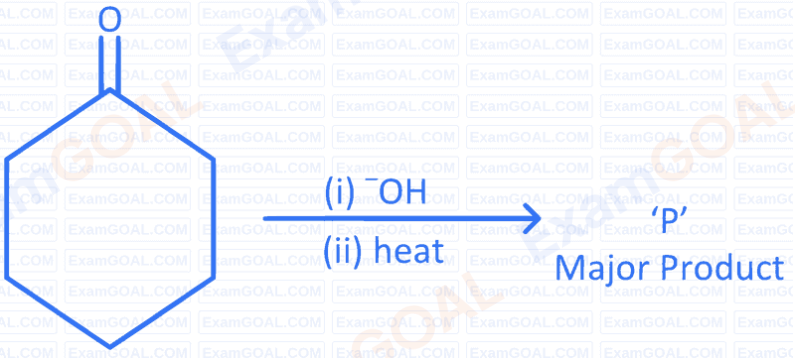
Open solution

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Q23. Numerical

Open solution

In the given reaction,



The number of π electrons present in the product 'P' is _____.

Mathematics 20 questions

Q24. MCQ (Single Correct Answer)

Open solution

Let α be a root of the equation $1 + x^2 + x^4 = 0$. Then, the value of $\alpha^{1011} + \alpha^{2022} - \alpha^{3033}$ is equal to :

(A) 1 (B) α (C) $1 + \alpha$ (D) $1 + 2\alpha$

Q25. MCQ (Single Correct Answer)

Open solution

Let arg(z) represent the principal argument of the complex number z. Then, $|z| = 3$ and $\arg(z - 1) - \arg(z + 1) = \frac{\pi}{4}$ intersect :

(A) exactly at one point. (B) exactly at two points.

(C) nowhere. (D) at infinitely many points.

Q26. MCQ (Single Correct Answer)

Open solution

The value of $\lim\limits_{x\to 1} \frac{(x^2-1)\sin^2(\pi x)}{x^4-2x^3+2x-1}$ is equal to:

(A) $\frac{\pi^2}{6}$ (B) $\frac{\pi^2}{3}$ (C) $\frac{\pi^2}{2}$ (D) π^2

Q27. MCQ (Single Correct Answer)

Open solution

Let f be a real valued continuous function on [0, 1] and $f(x) = x + \int_0^1 (x - t)f(t)dt$.

Then, which of the following points (x, y) lies on the curve y = f(x) ?

(A) (2, 4) (B) (1, 2) (C) (4, 17) (D) (6, 8)

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Q28. MCQ (Single Correct Answer)

If $\int_0^2 \left(\sqrt{2x} - \sqrt{2x - x^2}\right) dx = \int_0^1 \left(1 - \sqrt{1 - y^2} - \frac{y^2}{2}\right) dy + \int_1^2 \left(2 - \frac{y^2}{2}\right) dy + I$, then I equals

(A) $\int_0^1 \left(1 + \sqrt{1 - y^2}\right) dy$

(B) $\int_0^1 \left(\frac{y^2}{2} - \sqrt{1 - y^2} + 1\right) dy$

(C) $\int_0^1 \left(1 - \sqrt{1 - y^2}\right) dy$

(D) $\int_0^1 \left(\frac{y^2}{2} + \sqrt{1 - y^2} + 1\right) dy$

Open solution

Q29. MCQ (Single Correct Answer)

If y = y(x) is the solution of the differential equation $\left(1 + e^{2x}\right) \frac{dy}{dx} + 2\left(1 + y^2\right) e^x = 0$ and y (0) = 0, then $6\left(y'(0) + \left(y\left(\log_e \sqrt{3}\right)\right)^2\right)$ is equal to

(A) 2 (B) -2 (C) -4 (D) -1

Open solution

Q30. MCQ (Single Correct Answer)

Let a triangle ABC be inscribed in the circle $x^2 - \sqrt{2}(x + y) + y^2 = 0$ such that $\angle BAC = \frac{\pi}{2}$. If the length of side AB is $\sqrt{2}$, then the area of the ΔABC is equal to :

(A) 1

(B) $\left(\sqrt{6} + \sqrt{3}\right)/2$

(C) $\left(3 + \sqrt{3}\right)/4$

(D) $\left(\sqrt{6} + 2\sqrt{3}\right)/4$

Open solution

Q31. MCQ (Single Correct Answer)

The distance of the origin from the centroid of the triangle whose two sides have the equations $x - 2y + 1 = 0$ and $2x - y - 1 = 0$ and whose orthocenter is $\left(\frac{7}{3}, \frac{7}{3}\right)$ is :

(A) $\sqrt{2}$ (B) 2 (C) $2\sqrt{2}$ (D) 4

Open solution

Q32. MCQ (Single Correct Answer)

Let A, B, C be three points whose position vectors respectively are

$\vec{a} = \hat{i} + 4\hat{j} + 3\hat{k}$

$\vec{b} = 2\hat{i} + \alpha\hat{j} + 4\hat{k}$, $\alpha \in R$

$\vec{c} = 3\hat{i} - 2\hat{j} + 5\hat{k}$

If α is the smallest positive integer for which $\vec{a}, \vec{b}, \vec{c}$ are noncollinear, then the length of the median, in ΔABC, through A is :

(A) $\frac{\sqrt{82}}{2}$ (B) $\frac{\sqrt{62}}{2}$ (C) $\frac{\sqrt{69}}{2}$ (D) $\frac{\sqrt{66}}{2}$

Open solution

Q33. MCQ (Single Correct Answer)

The probability that a relation R from {x, y} to {x, y} is both symmetric and transitive, is equal to :

(A) $\frac{5}{16}$ (B) $\frac{9}{16}$ (C) $\frac{11}{16}$ (D) $\frac{13}{16}$

Open solution

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Q34. MCQ (Single Correct Answer)

The number of values of a $\in$ N such that the variance of 3, 7, 12, a, 43 $-$ a is a natural number is :

(A) 0 (B) 2 (C) 5 (D) infinite

Open solution

Q35. Numerical

Let $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} + \left(\vec{b} \times \vec{c}\right) = \vec{0}$ and $\vec{b} . \vec{c} = 5$. Then the value of $3\left(\vec{c} . \vec{a}\right)$ is equal to _____.

Open solution

Q36. Numerical

Let y = y(x), x > 1, be the solution of the differential equation $(x-1)\frac{dy}{dx} + 2xy = \frac{1}{x-1}$, with $y(2) = \frac{1+e^4}{2e^4}$. If $y(3) = \frac{e^\alpha+1}{\beta e^\alpha}$, then the value of $\alpha + \beta$ is equal to _____.

Open solution

Q37. Numerical

Let 3, 6, 9, 12, ... upto 78 terms and 5, 9, 13, 17, ... upto 59 terms be two series. Then, the sum of the terms common to both the series is equal to _____.

Open solution

Q38. Numerical

For real numbers a, b (a > b > 0), let

Area $\left\{(x, y) : x^2 + y^2 \leq a^2 \text{ and } \frac{x^2}{a^2} + \frac{y^2}{b^2} \geq 1\right\} = 30\pi$

and

Area $\left\{(x, y) : x^2 + y^2 \geq b^2 \text{ and } \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1\right\} = 18\pi$

Then, the value of $(a - b)^2$ is equal to _____.

Open solution

Q39. Numerical

Let f and g be twice differentiable even functions on $(-2, 2)$ such that $f\left(\frac{1}{4}\right) = 0$, $f\left(\frac{1}{2}\right) = 0$, $f(1) = 1$ and $g\left(\frac{3}{4}\right) = 0$, $g(1) = 2$. Then, the minimum number of solutions of $f(x)g''(x) + f'(x)g'(x) = 0$ in $(-2, 2)$ is equal to _____.

Open solution

Q40. Numerical

Let the coefficients of x^{-1} and x^{-3} in the expansion of $\left(2x^{\frac{1}{5}} - \frac{1}{x^{\frac{1}{5}}}\right)^{15}$, $x > 0$, be m and n respectively. If r is a positive integer such that $mn^2 = {}^{15}C_r \cdot 2^r$, then the value of r is equal to _____.

Open solution

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Q41. Numerical

The total number of four digit numbers such that each of first three digits is divisible by the last digit, is equal to _____.

Open solution

Q42. Numerical

Let $M = \begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}$, where α is a non-zero real number an $N = \sum_{k=1}^{49} M^{2k}$. If $(I - M^2)N = -2I$, then the positive integral value of α is _____.

Open solution

Q43. Numerical

Let f(x) and g(x) be two real polynomials of degree 2 and 1 respectively. If $f(g(x)) = 8x^2 - 2x$ and $g(f(x)) = 4x^2 + 6x + 1$, then the value of $f(2) + g(2)$ is _____.

Open solution

Physics 27 questions

Q44. MCQ (Single Correct Answer)

A small toy starts moving from the position of rest under a constant acceleration. If it travels a distance of 10m in t s, the distance travelled by the toy in the next t s will be :

(A) 10 m (B) 20 m (C) 30 m (D) 40 m

Open solution

Q45. MCQ (Single Correct Answer)

At what temperature a gold ring of diameter 6.230 cm be heated so that it can be fitted on a wooden bangle of diameter 6.241 cm ? Both the diameters have been measured at room temperature (27°C).

(Given : coefficient of linear thermal expansion of gold $\alpha_L = 1.4 \times 10^{-5}\ K^{-1}$)

(A) 125.7°C (B) 91.7°C (C) 425.7°C (D) 152.7°C

Open solution

Q46. MCQ (Single Correct Answer)

Two point charges Q each are placed at a distance d apart. A third point charge q is placed at a distance x from mid-point on the perpendicular bisector. The value of x at which charge q will experience the maximum Coulomb's force is :

(A) x = d (B) $x = \frac{d}{2}$ (C) $x = \frac{d}{\sqrt{2}}$ (D) $x = \frac{d}{2\sqrt{2}}$

Open solution

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Q47. MCQ (Single Correct Answer)

The speed of light in media 'A' and 'B' are 2.0×10^{10} cm/s and 1.5×10^{10} cm/s respectively. A ray of light enters from the medium B to A at an incident angle 'θ'. If the ray suffers total internal reflection, then

(A) $\theta = \sin^{-1}\left(\frac{3}{4}\right)$ (B) $\theta > \sin^{-1}\left(\frac{2}{3}\right)$

(C) $\theta < \sin^{-1}\left(\frac{3}{4}\right)$ (D) $\theta > \sin^{-1}\left(\frac{3}{4}\right)$

Open solution

Q48. MCQ (Single Correct Answer)

The electric field at a point associated with a light wave is given by

E = 200 [sin (6×10^{15})t + sin (9×10^{15})t] Vm^{-1}

Given : h = 4.14 $\times 10^{-15}$ eVs

If this light falls on a metal surface having a work function of 2.50 eV, the maximum kinetic energy of the photoelectrons will be

(A) 1.90 eV (B) 3.27 eV (C) 3.60 eV (D) 3.42 eV

Open solution

Q49. MCQ (Single Correct Answer)

A capacitor is discharging through a resistor R. Consider in time t_1, the energy stored in the capacitor reduces to half of its initial value and in time t_2, the charge stored reduces to one eighth of its initial value. The ratio t_1/t_2 will be

(A) 1/2 (B) 1/3 (C) 1/4 (D) 1/6

Open solution

Q50. MCQ (Single Correct Answer)

Starting with the same initial conditions, an ideal gas expands from volume V_1 to V_2 in three different ways. The work done by the gas is W_1 if the process is purely isothermal, W_2, if the process is purely adiabatic and W_3 if the process is purely isobaric. Then, choose the correct option

(A) $W_1 < W_2 < W_3$ (B) $W_2 < W_3 < W_1$ (C) $W_3 < W_1 < W_2$ (D) $W_2 < W_1 < W_3$

Open solution

Q51. MCQ (Single Correct Answer)

Two long current carrying conductors are placed parallel to each other at a distance of 8 cm between them. The magnitude of magnetic field produced at mid-point between the two conductors due to current flowing in them is 300 μT. The equal current flowing in the two conductors is :

(A) 30A in the same direction. (B) 30A in the opposite direction.

(C) 60A in the opposite direction. (D) 300A in the opposite direction.

Open solution

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Q52. MCQ (Single Correct Answer)

The time period of a satellite revolving around earth in a given orbit is 7 hours. If the radius of orbit is increased to three times its previous value, then approximate new time period of the satellite will be

Open solution

(A) 40 hours (B) 36 hours (C) 30 hours (D) 25 hours

Q53. MCQ (Single Correct Answer)

The motion of a simple pendulum executing S.H.M. is represented by the following equation.

$y = A\sin(\pi t + \phi)$, where time is measured in second. The length of pendulum is

Open solution

(A) 97.23 cm (B) 25.3 cm (C) 99.4 cm (D) 406.1 cm

Q54. MCQ (Single Correct Answer)

A vessel contains 16g of hydrogen and 128g of oxygen at standard temperature and pressure. The volume of the vessel in cm^3 is :

Open solution

(A) 72×10^5 (B) 32×10^5 (C) 27×10^4 (D) 54×10^4

Q55. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : The electric force changes the speed of the charged particle and hence changes its kinetic energy; whereas the magnetic force does not change the kinetic energy of the charged particle.

Statement II : The electric force accelerates the positively charged particle perpendicular to the direction of electric field. The magnetic force accelerates the moving charged particle along the direction of magnetic field.

In the light of the above statements, choose the most appropriate answer from the options given below :

Open solution

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

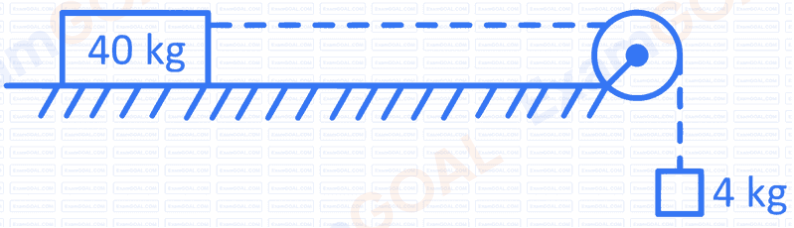
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Q56. MCQ (Single Correct Answer)

Open solution

A block of mass 40 kg slides over a surface, when a mass of 4 kg is suspended through an inextensible massless string passing over frictionless pulley as shown below.

The coefficient of kinetic friction between the surface and block is 0.02. The acceleration of block is. (Given g = 10 ms^{-2}.)

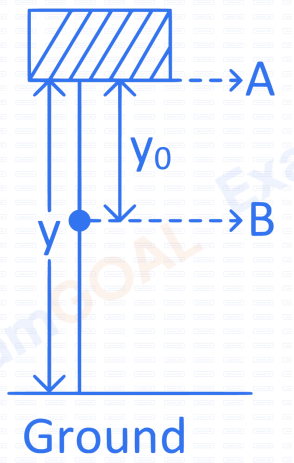


(A) 1 ms^{-2} (B) 1/5 ms^{-2} (C) 4/5 ms^{-2} (D) 8/11 ms^{-2}

Q57. MCQ (Single Correct Answer)

Open solution

In the given figure, the block of mass m is dropped from the point 'A'. The expression for kinetic energy of block when it reaches point 'B' is



(A) $\frac{1}{2}mg\,{y_0}^2$ (B) $\frac{1}{2}mg\,y^2$ (C) $mg(y-y_0)$ (D) mgy_0

Q58. MCQ (Single Correct Answer)

Open solution

A block of mass M placed inside a box descends vertically with acceleration 'a'. The block exerts a force equal to one-fourth of its weight on the floor of the box. The value of 'a' will be

(A) $\frac{g}{4}$ (B) $\frac{g}{2}$ (C) $\frac{3g}{4}$ (D) g

Q59. MCQ (Single Correct Answer)

Open solution

If the electric potential at any point (x, y, z) m in space is given by V = $3x^2$ volt. The electric field at the point (1, 0, 3) m will be :

(A) 3 Vm^{-1}, directed along positive x-axis.
(B) 3 Vm^{-1}, directed along negative x-axis.
(C) 6 Vm^{-1}, directed along positive x-axis.
(D) 6 Vm^{-1}, directed along negative x-axis.

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Q60. MCQ (Single Correct Answer)

The combination of two identical cells, whether connected in series or parallel combination provides the same current through an external resistance of 2Ω. The value of internal resistance of each cell is

(A) 2Ω (B) 4Ω (C) 6Ω (D) 8Ω

Open solution

Q61. MCQ (Single Correct Answer)

A person can throw a ball upto a maximum range of 100 m. How high above the ground he can throw the same ball?

(A) 25 m (B) 50 m (C) 100 m (D) 200 m

Open solution

Q62. Numerical

The Vernier constant of Vernier callipers is 0.1 mm and it has zero error of (−0.05) cm. While measuring diameter of a sphere, the main scale reading is 1.7 cm and coinciding vernier division is 5. The corrected diameter will be _____ $\times 10^{-2}$ cm.

Open solution

Q63. Numerical

A small spherical ball of radius 0.1 mm and density 10^4 kg m^{-3} falls freely under gravity through a distance h before entering a tank of water. If, after entering the water the velocity of ball does not change and it continue to fall with same constant velocity inside water, then the value of h will be _____ m.

(Given g = 10 ms^{-2}, viscosity of water = 1.0×10^{-5} N-sm^{-2}).

Open solution

Q64. Numerical

In an experiment to determine the velocity of sound in air at room temperature using a resonance tube, the first resonance is observed when the air column has a length of 20.0 cm for a tuning fork of frequency 400 Hz is used. The velocity of the sound at room temperature is 336 ms^{-1}. The third resonance is observed when the air column has a length of _____ cm.

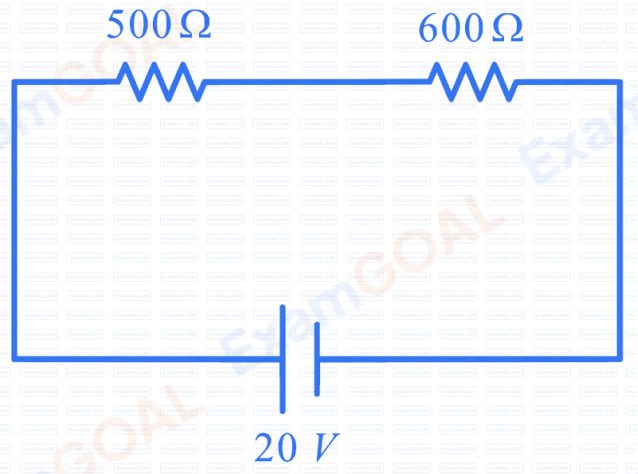
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Q65. Numerical

Two resistors are connected in series across a battery as shown in figure. If a voltmeter of resistance 2000 Ω is used to measure the potential difference across 500 Ω resistor, the reading of the voltmeter will be _____ V.

Open solution



Q66. Numerical

A potential barrier of 0.4 V exists across a p-n junction. An electron enters the junction from the n-side with a speed of 6.0×10^5 ms^{-1}. The speed with which electron enters the p side will be $\frac{x}{3} \times 10^5$ ms^{-1} the value of x is _____.

(Given mass of electron = 9×10^{-31} kg, charge on electron = 1.6×10^{-19} C.)

Open solution

Q67. Numerical

The displacement current of 4.425 μA is developed in the space between the plates of parallel plate capacitor when voltage is changing at a rate of 10^6 Vs^{-1}. The area of each plate of the capacitor is 40 cm^2. The distance between each plate of the capacitor is $x \times 10^{-3}$ m. The value of x is _____.

(Permittivity of free space, ε_0 = 8.85×10^{-12} C^2 N^{-1} m^{-2}).

Open solution

Q68. Numerical

The moment of inertia of a uniform thin rod about a perpendicular axis passing through one end is I_1. The same rod is bent into a ring and its moment of inertia about a diameter is I_2. If $\frac{I_1}{I_2}$ is $\frac{x\pi^2}{3}$, then the value of x will be _____.

Open solution

Q69. Numerical

In a double slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-3} cm. If the distance between the slits is 1 mm, then the wavelength of the light will be _____ nm.

Open solution

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Q70. Numerical

An inductor of 0.5 mH, a capacitor of 200 μF and a resistor of 2 Ω are connected in series with a 220 V ac source. If the current is in phase with the emf, the frequency of ac source will be _____ $\times 10^2$ Hz.

Open solution

Out of syllabus 20 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q71. MCQ (Single Correct Answer) · **Chemistry**

Using the rules for significant figures, the correct answer for the expression $\frac{0.02858 \times 0.112}{0.5702}$ will be

(A) 0.005613 **(B)** 0.00561 **(C)** 0.0056 **(D)** 0.006

Open solution

Q72. MCQ (Single Correct Answer) · **Chemistry**

A 42.12% (w/v) solution of NaCl causes precipitation of a certain sol in 10 hours. The coagulating value of NaCl for the sol in 2 hours is :

[Given : Molar mass : Na = 23.0 g mol^{-1} ; Cl = 35.5 g mol^{-1}]

(A) 36 m mol L^{-1} **(B)** 36 mol L^{-1} **(C)** 1440 mol L^{-1} **(D)** 1440 m mol L^{-1}

Open solution

Q73. MCQ (Single Correct Answer) · **Chemistry**

Match List I with List II.

	List-I Ore		List-II Composition
(A)	Siderite	(I)	$FeCO_3$
(B)	Malachite	(II)	$CuCO_3 . Cu(OH)_2$
(C)	Sphalerite	(III)	ZnS
(D)	Calamine	(IV)	$ZnCO_3$

Choose the correct answer from the options given below :

(A) A-I, B-II, C-III, D-IV **(B)** A-III, B-IV, C-II, D-I

(C) A-IV, B-III, C-I, D-II **(D)** A-I, B-II, C-IV, D-III

Open solution

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Q74. MCQ (Single Correct Answer) · **Chemistry**

Amongst baking soda, caustic soda and washing soda, carbonate anion is present in :

(A) washing soda only.

(B) washing soda and caustic soda only.

(C) washing soda and baking soda only.

(D) baking soda, caustic soda and washing soda.

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

Aqueous solution of which of the following boron compounds will be strongly basic in nature?

(A) $NaBH_4$ (B) $LiBH_4$ (C) B_2H_6 (D) $Na_2B_4O_7$

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Dacron is an example of polyester polymer.

Reason R : Dacron is made up of ethylene glycol and terephthalic acid monomers.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both A and R are correct and R is the correct explanation of A.

(B) Both A and R are correct but R is NOT the correct explanation of A.

(C) A is correct but R is not correct.

(D) A is not correct but R is correct.

Open solution

Q77. MCQ (Single Correct Answer) · **Chemistry**

The mixture of chloroxylenol and terpineol is an example of

(A) antiseptic (B) pesticide

(C) disinfectant (D) narcotic analgesic

Open solution

Q78. MCQ (Single Correct Answer) · **Mathematics**

Let $A = \begin{pmatrix} 2 & -1 \\ 0 & 2 \end{pmatrix}$. If $B = I - {}^5C_1(adj\ A) + {}^5C_2(adj\ A)^2 - \ldots\ - {}^5C_5(adj\ A)^5$, then the sum of all elements of the matrix B is

(A) -5 (B) -6 (C) -7 (D) -8

Open solution

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Q79. MCQ (Single Correct Answer) · **Mathematics**

The sum of the infinite series $1+\frac{5}{6}+\frac{12}{6^2}+\frac{22}{6^3}+\frac{35}{6^4}+\frac{51}{6^5}+\frac{70}{6^6}+\ldots\ldots$ is equal to :

(A) $\frac{425}{216}$ (B) $\frac{429}{216}$ (C) $\frac{288}{125}$ (D) $\frac{280}{125}$

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

Let f : R $\rightarrow$ R be a function defined by f(x) = $(x-3)^{n_1}(x-5)^{n_2}$, $n_1, n_2 \in N$. Then, which of the following is NOT true?

(A) For $n_1 = 3, n_2 = 4$, there exists $\alpha \in (3, 5)$ where f attains local maxima.

(B) For $n_1 = 4, n_2 = 3$, there exists $\alpha \in (3, 5)$ where f attains local minima.

(C) For $n_1 = 3, n_2 = 5$, there exists $\alpha \in (3, 5)$ where f attains local maxima.

(D) For $n_1 = 4, n_2 = 6$, there exists $\alpha \in (3, 5)$ where f attains local maxima.

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Let P : $y^2 = 4ax$, a > 0 be a parabola with focus S. Let the tangents to the parabola P make an angle of $\frac{\pi}{4}$ with the line y = 3x + 5 touch the parabola P at A and B. Then the value of a for which A, B and S are collinear is :

(A) 8 only (B) 2 only (C) $\frac{1}{4}$ only (D) any a > 0

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let $\frac{x-2}{3}=\frac{y+1}{-2}=\frac{z+3}{-1}$ lie on the plane $px-qy+z=5$, for some p, q $\in$ R. The shortest distance of the plane from the origin is :

(A) $\sqrt{\frac{3}{109}}$ (B) $\sqrt{\frac{5}{142}}$ (C) $\frac{5}{\sqrt{71}}$ (D) $\frac{1}{\sqrt{142}}$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Let Q be the mirror image of the point P(1, 2, 1) with respect to the plane x + 2y + 2z = 16. Let T be a plane passing through the point Q and contains the line $\vec{r}=-\hat{k}+\lambda\left(\hat{i}+\hat{j}+2\hat{k}\right)$, $\lambda \in R$. Then, which of the following points lies on T?

(A) (2, 1, 0) (B) (1, 2, 1) (C) (1, 2, 2) (D) (1, 3, 2)

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

From the base of a pole of height 20 meter, the angle of elevation of the top of a tower is 60°. The pole subtends an angle 30° at the top of the tower. Then the height of the tower is :

(A) $15\sqrt{3}$ (B) $20\sqrt{3}$ (C) $20 + 10\sqrt{3}$ (D) 30

Open solution

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Q85. MCQ (Single Correct Answer) · **Mathematics**

Negation of the Boolean statement $(p \vee q) \Rightarrow ((\sim r) \vee p)$ is equivalent to :

(A) $p \wedge (\sim q) \wedge r$

(B) $(\sim p) \wedge (\sim q) \wedge r$

(C) $(\sim p) \wedge q \wedge r$

(D) $p \wedge q \wedge (\sim r)$

Open solution

Q86. MCQ (Single Correct Answer) · **Mathematics**

Let $n \geq 5$ be an integer. If $9^n - 8n - 1 = 64\alpha$ and $6^n - 5n - 1 = 25\beta$, then $\alpha - \beta$ is equal to

(A) $1 + {}^nC_2(8-5) + {}^nC_3(8^2-5^2) + \ldots\ldots + {}^nC_n(8^{n-1}-5^{n-1})$

(B) $1 + {}^nC_3(8-5) + {}^nC_4(8^2-5^2) + \ldots\ldots + {}^nC_n(8^{n-2}-5^{n-2})$

(C) ${}^nC_3(8-5) + {}^nC_4(8^2-5^2) + \ldots\ldots + {}^nC_n(8^{n-2}-5^{n-2})$

(D) ${}^nC_4(8-5) + {}^nC_5(8^2-5^2) + \ldots\ldots + {}^nC_n(8^{n-3}-5^{n-3})$

Open solution

Q87. Numerical · **Mathematics**

The number of solutions of the equation $\sin x = \cos^2 x$ in the interval (0, 10) is _____.

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

In the following nuclear reaction,

$D \xrightarrow{\alpha} D_1 \xrightarrow{\beta^-} D_2 \xrightarrow{\alpha} D_3 \xrightarrow{\gamma} D_4$

Mass number of D is 182 and atomic number is 74. Mass number and atomic number of D_4 respectively will be _____.

(A) 174 and 71

(B) 174 and 69

(C) 172 and 69

(D) 172 and 71

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

The TV transmission tower at a particular station has a height of 125 m. For doubling the coverage of its range, the height of the tower should be increased by

(A) 125 m

(B) 250 m

(C) 375 m

(D) 500 m

Open solution

Q90. Numerical · **Physics**

The half life of a radioactive substance is 5 years. After x years a given sample of the radioactive substance gets reduced to 6.25% of its initial value. The value of x is _____.

Open solution

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Answer Key

Chemistry

1. **B**	2. **B**	3. **A**	4. **B**	5. **C**
6. **C**	7. **D**	8. **D**	9. **C**	10. **C**
11. **D**	12. **C**	13. **B**	14. **29**	15. **195**
16. **3**	17. **7**	18. **216**	19. **6**	20. **5**
21. **2**	22. **8**	23. **4**		

Mathematics

24. **A**	25. **C**	26. **D**	27. **D**	28. **C**
29. **C**	30. **A**	31. **C**	32. **A**	33. **A**
34. **A**	35. **BONUS**	36. **14**	37. **2223**	38. **12**
39. **4**	40. **5**	41. **1086**	42. **1**	43. **18**

Physics

44. **C**	45. **D**	46. **D**	47. **D**	48. **D**
49. **D**	50. **D**	51. **B**	52. **B**	53. **C**
54. **C**	55. **C**	56. **D**	57. **D**	58. **C**
59. **D**	60. **A**	61. **B**	62. **180**	63. **20**
64. **104**	65. **8**	66. **14**	67. **8**	68. **8**
69. **600**	70. **5**			

Out of syllabus

71. **B**	72. **D**	73. **A**	74. **A**	75. **D**
76. **A**	77. **A**	78. **C**	79. **C**	80. **C**
81. **D**	82. **B**	83. **B**	84. **D**	85. **C**
86. **C**	87. **4**	88. **A**	89. **C**	90. **20**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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